

Data

Where the training mixture comes from, and how anyone knows it is good

Required

Penedo et al., The FineWeb Datasets · [arXiv:2406.17557](#)

Optional

Li et al., DataComp-LM · [arXiv:2406.11794](#) Xie et al., DoReMi · [arXiv:2305.10429](#) Wang et al., Self-Instruct · [arXiv:2212.10560](#)
Shumailov et al., The Curse of Recursion · [arXiv:2305.17493](#) Gunasekar et al., Textbooks Are All You Need · [arXiv:2306.11644](#)

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Quick overview

Where Thursday stopped. The judge's biases measured, contamination, error bars, the harness.

Then the levers, and lever one is today

The mixture. What a model eats, and when the human-written web runs out.

The mixture is a choice

The pipeline. FineWeb, stage by stage, with your vote on four real documents.

Dedup, filters, a learned classifier

Synthesis. You already trained on synthetic text. Then collapse, on your codebase.

TinyStories, textbooks, Self-Instruct, recursion

The meter. Who grades the data: humans, judges, classifiers.

What each costs, what each cannot see

1 WHERE THURSDAY STOPPED

Six Ways a Number Lies

Failure Modes

The judge's

Position bias. Swap A and B, and the verdict changes.

Verbosity bias. Longer scores higher, content held equal.

Self-preference. The judge favors text in its own voice.

The benchmark's

Contamination. The test set leaked into the training data.

Format fragility. Reformat the questions, and the scores move.

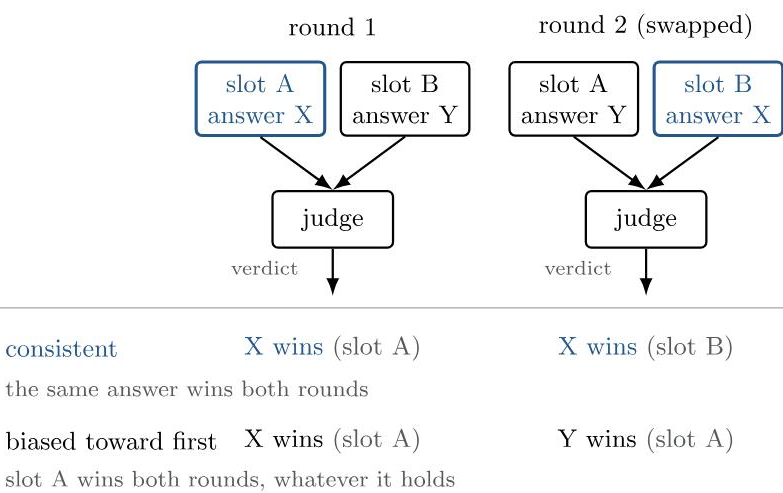
Saturation. Scores crowd the ceiling; the gaps left are noise.

Judging the Judge

Claude v1, 23.8: below random; the order IS the verdict

Judge	Prompt	Consistency	Biased toward first	Biased toward second	Error
Claude-v1	default	23.8%	75.0%	0.0%	1.2%
	rename	56.2%	11.2%	28.7%	3.8%
GPT-3.5	default	46.2%	50.0%	1.2%	2.5%
	rename	51.2%	38.8%	6.2%	3.8%
GPT-4	default	65.0%	30.0%	5.0%	0.0%
	rename	66.2%	28.7%	5.0%	0.0%

GPT-4: same winner after the swap, 65% of pairs
judged twice, A and B trading places; random verdicts agree 33%



default and **rename** are two judge prompts. The default prompt labels the two answers Assistant A and Assistant B; rename renames the assistants in that prompt, to see whether the bias is on positions or names.

consistent: the same answer wins both rounds after A and B trade places. Random verdicts (A, B, or tie, twice) agree 33% of the time, the paper's own random floor.

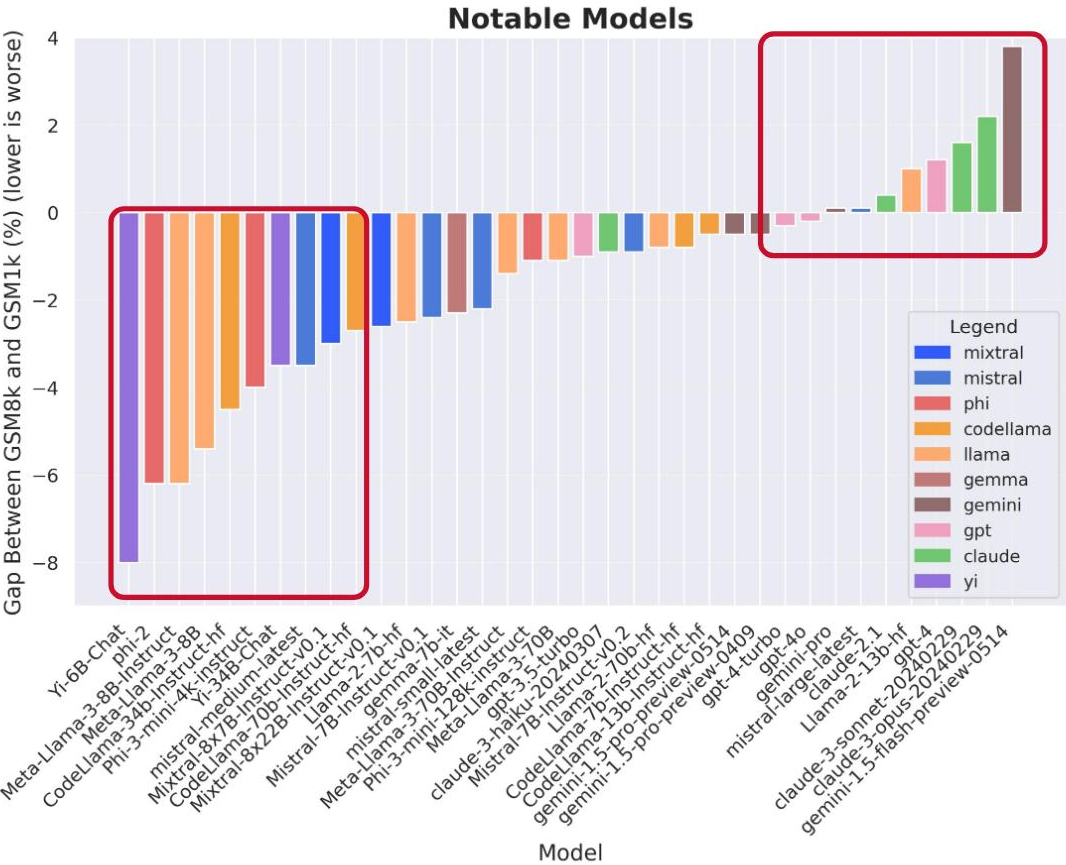
Judge	Claude-v1	GPT-3.5	GPT-4
Failure rate	91.3%	91.3%	8.7%

fooled by the padded list: Claude v1 91.3%, GPT-3.5 91.3%, GPT-4 8.7%
Table 3: 23 numbered-list answers, each list rephrased and prepended

Contamination

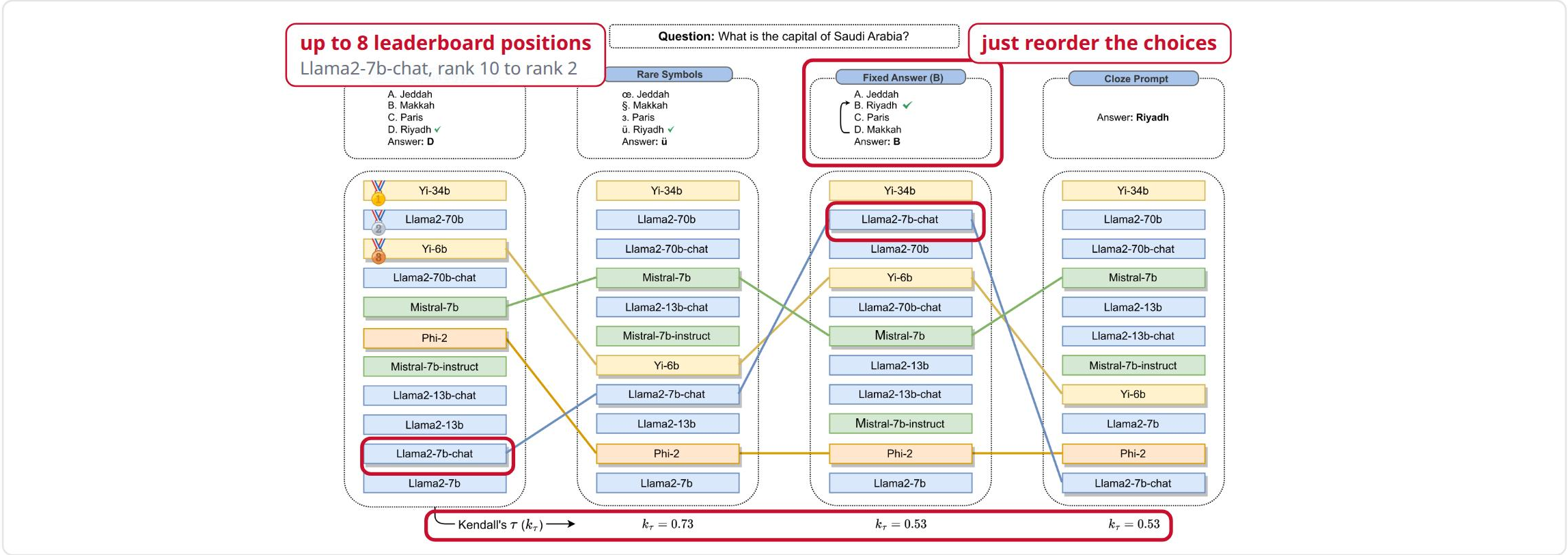
the setup: write 1,250 NEW problems at GSM8k difficulty; run every model on both tests; each bar = old score minus new score

Phi and Mistral families: up to 8 points lower on the fresh problems that gap is memorized test data, not math



frontier models: bar near zero, the skill is real

Fragile Rankings



Kendall's tau: agreement between two rankings; pairs of models kept in the same order minus pairs that flipped, over all pairs, so 1 is the same ranking, 0 unrelated, -1 reversed; the figure prints $k_\tau = (\tau + 1)/2$, the share of pairs still in order

11 models, 55 pairs: $k_\tau = 0.73$ means 40 pairs kept their order and 15 flipped; 0.53 means 29 kept and 26 flipped, just above the 0.5 that two unrelated rankings would give

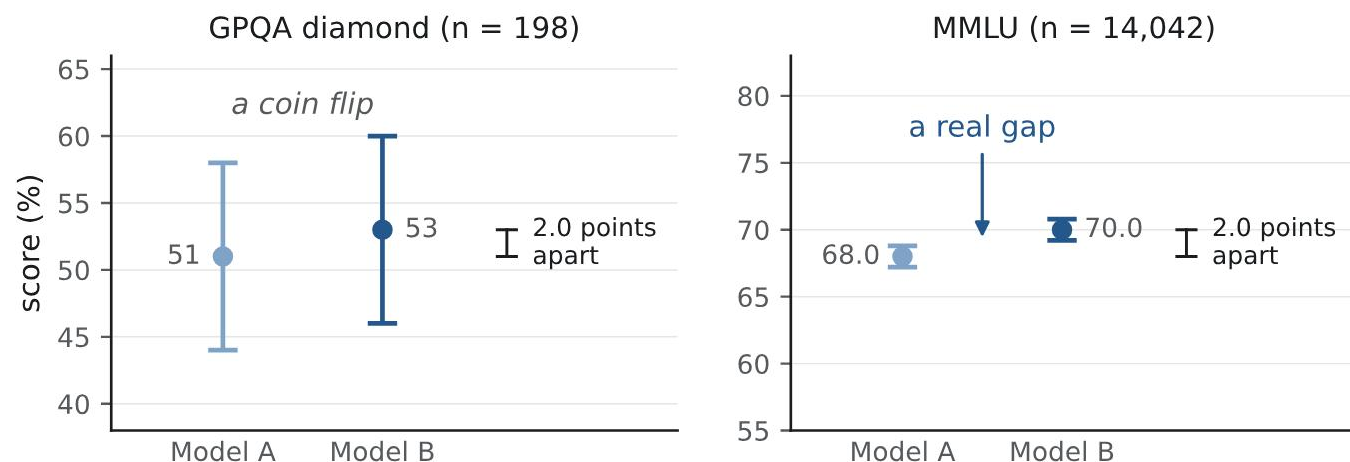
Error Bars

A benchmark score is an average over n questions, so it carries a standard error.

$$SE = \sqrt{\frac{p(1-p)}{n}}$$

MMLU, 14,042 questions: $\pm 1.96 \sqrt{\frac{0.7 \cdot 0.3}{14,042}} = 0.8$ percentage points

GPQA diamond, 198 questions: $\pm 1.96 \sqrt{\frac{0.5 \cdot 0.5}{198}} = 7.0$ percentage points



A two-point lead on a small benchmark is a coin flip. Paired, per-question comparison tightens it.

2 THE HARNESS

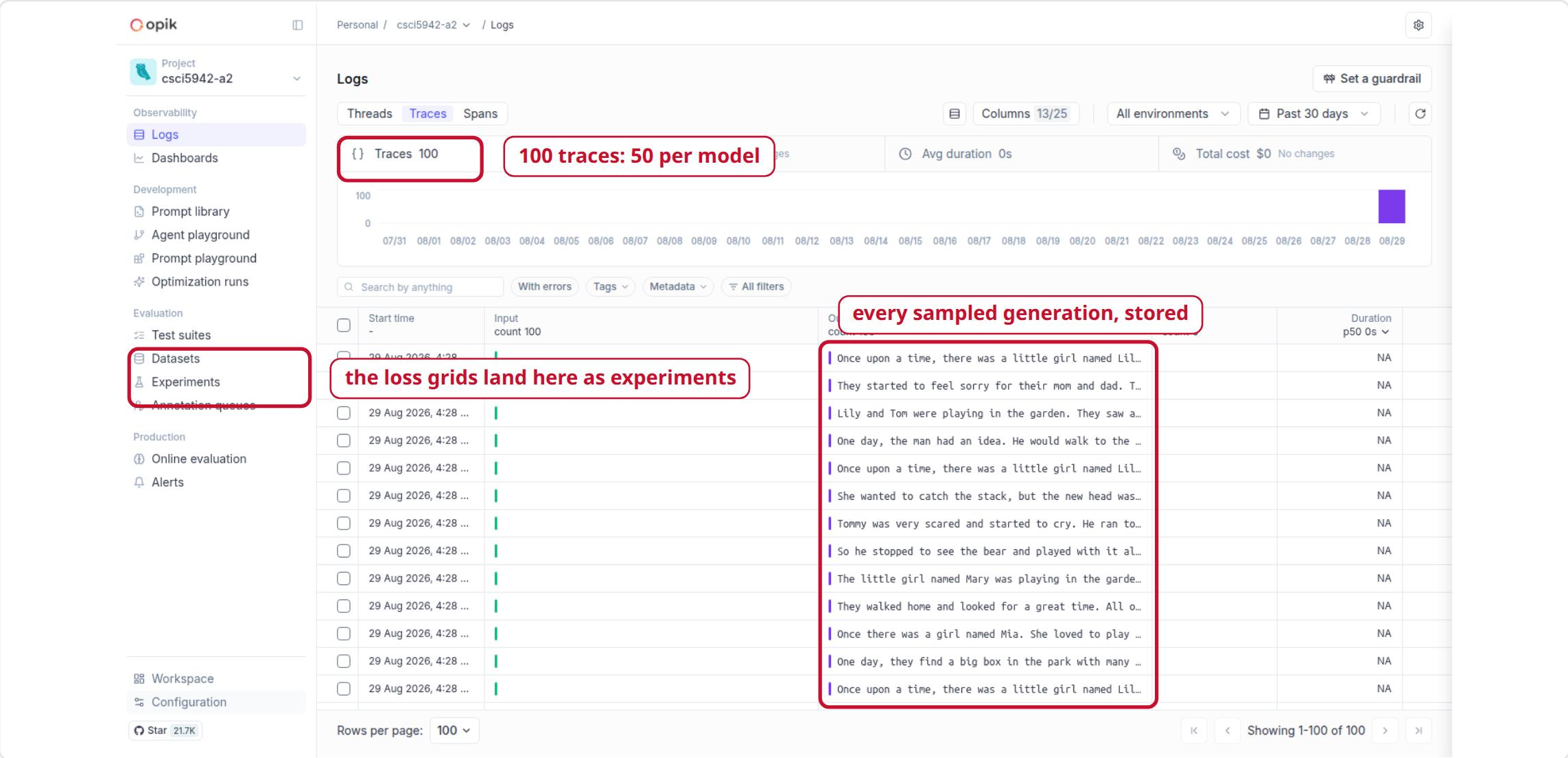
Evaluation as Infrastructure

Harnesses

```
task: mmlu_abstract_algebra
dataset_path: cais/mmlu
num_fewshot: 5
doc_to_choice: ["A", "B", "C", "D"]
metric: acc
```

The prompt, the shots, the scoring and the aggregation are pinned in config, or the number is not comparable. The same LLaMA 65B weights scored 63.7 on MMLU in one harness and 48.8 in another.

Opik



Self-hosted Opik, project csci5942-a2: the Assignment 1 base model and its TinyStories twin, logged by the A2 harness.

Assignment 2

Train the base config, unchanged, on a second corpus: TinyStories.

Same 82M characters of budget, two minutes on your A1 GPU

Evaluate both models on four domains: the loss grid from this lecture, yours.

Shakespeare, TinyStories, Wikipedia, Python, with the OOV accounting

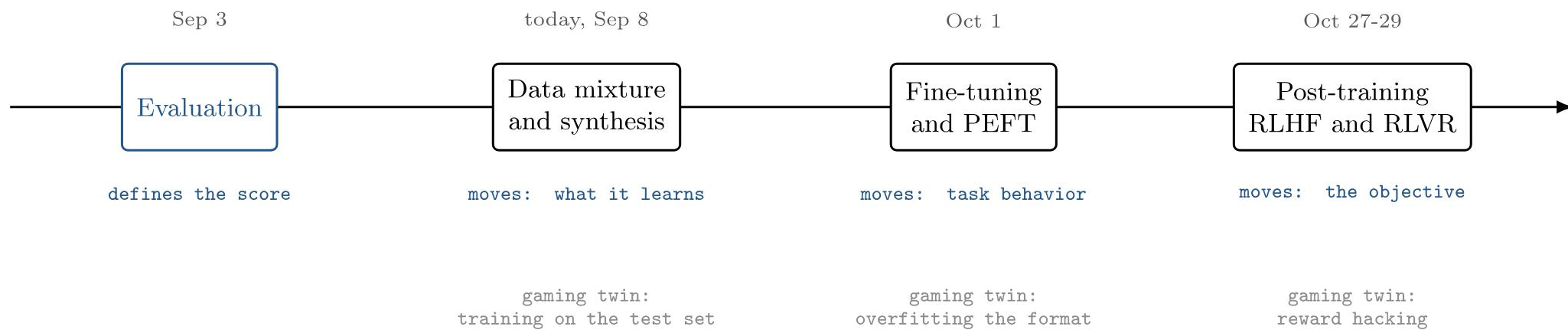
Judge both models' generations in Opik, with the order-swap control.

Gemini as judge, per-corpus rubric, error bars required

Deliverable: which model is better? Defend the answer per exam.

The correct answer names the exam it is true on

Making the Number Move

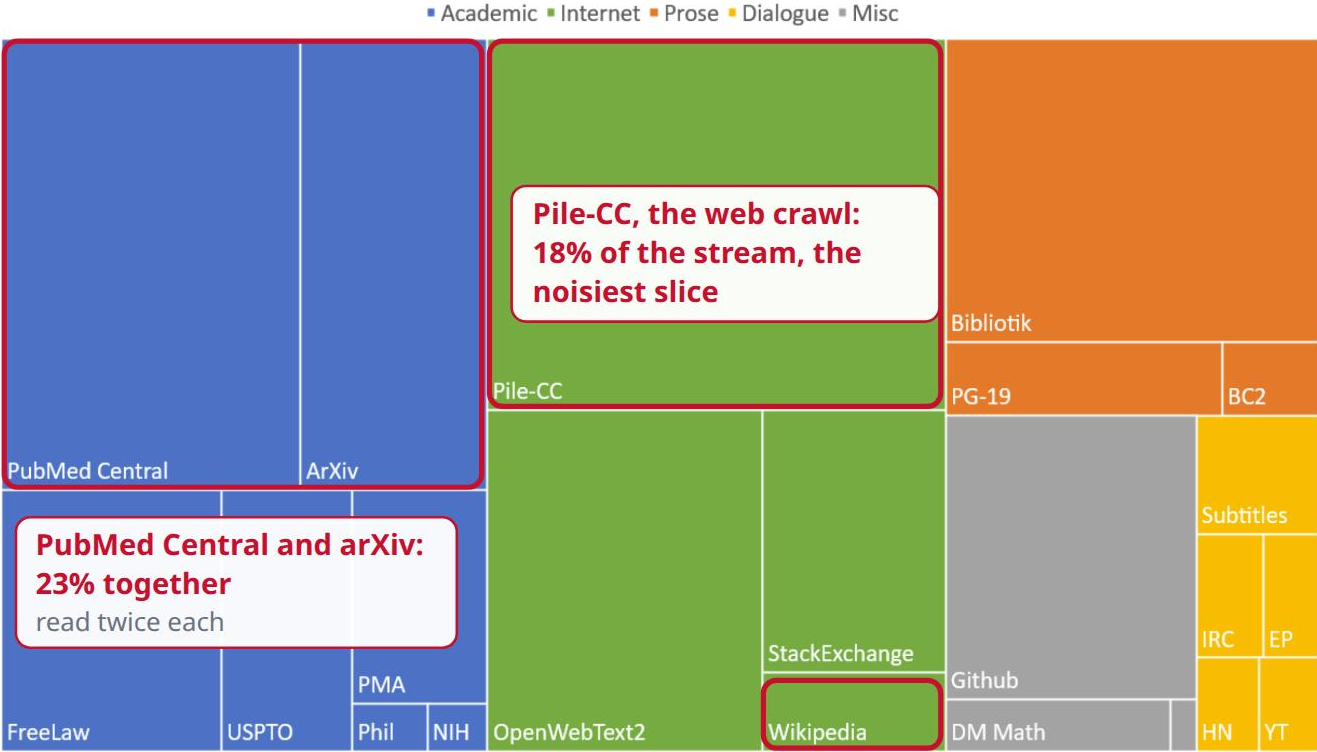


3 THE LEVER

The Mixture Decides

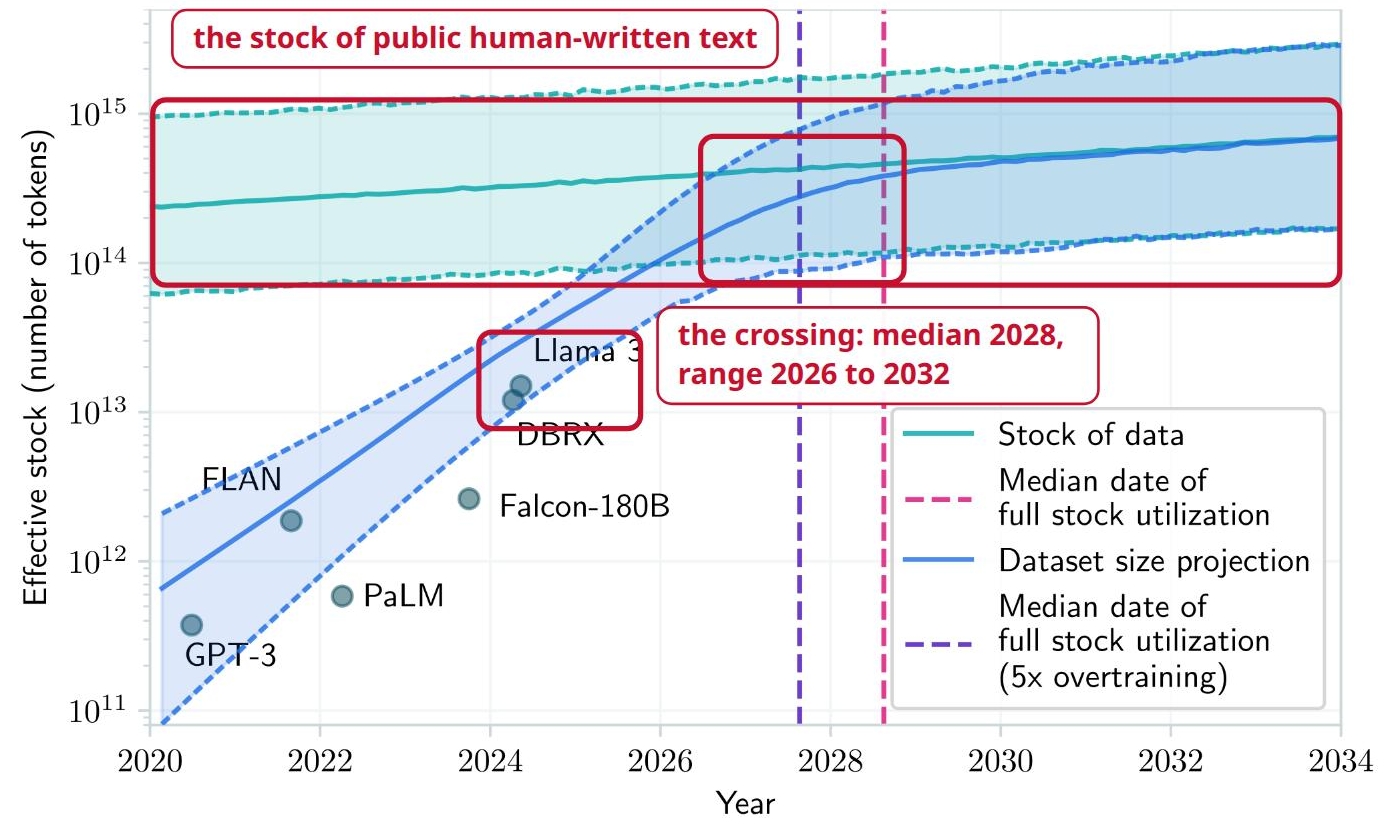
What a Model Eats

Composition of the Pile by Category



$C \xrightarrow{\text{power law}} L_{\text{mixture}} \xrightarrow{\text{link}} \text{score}$ Compute buys loss on the mixture, and the link turns that loss into a score.

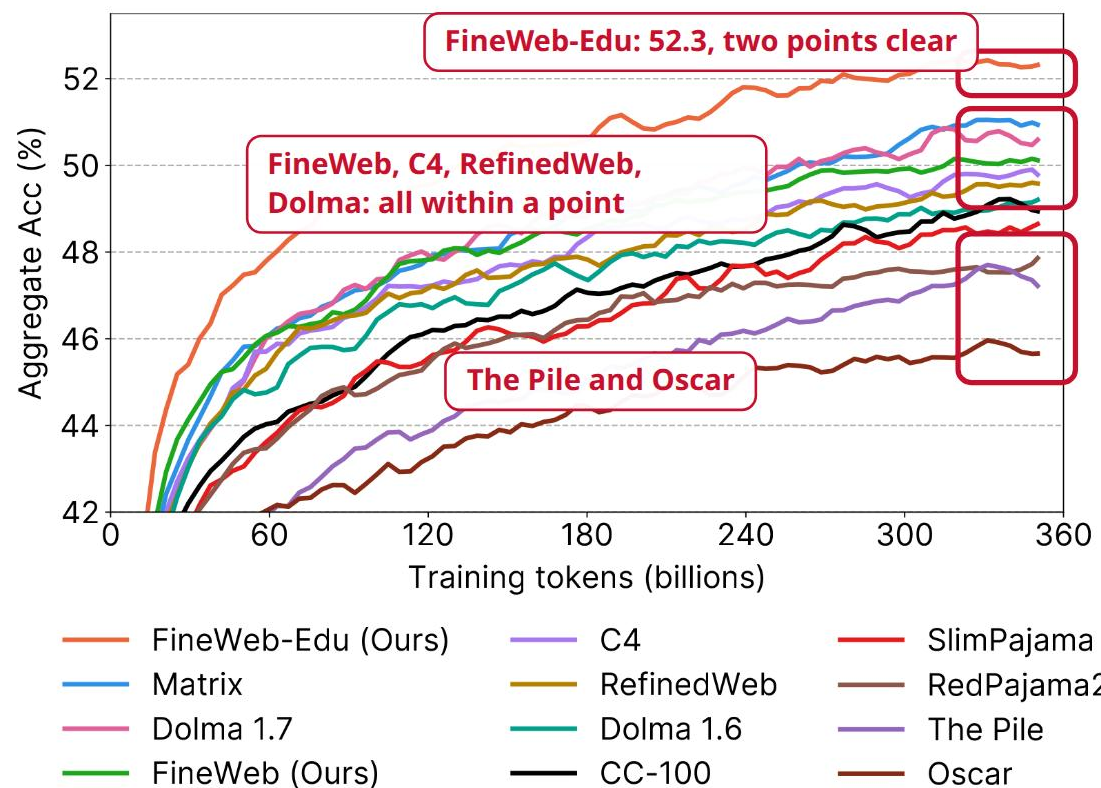
Running Out



one frontier run: Llama 3 pretrained on 15T tokens, one twentieth of the 320T effective stock

the exits: repeat what exists, keep only the good part, or write more

The Ablation



the instrument: 1.71B parameters, same architecture and tokenizer, a fixed token budget; only the training data changes

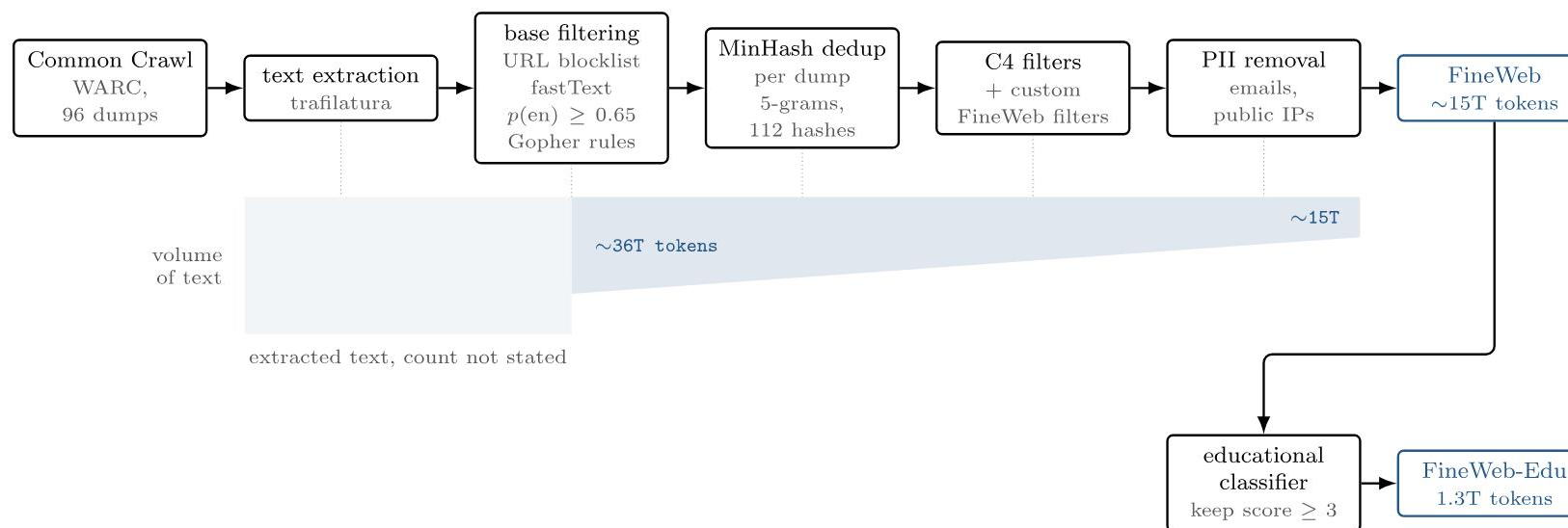
score: the mean over eight benchmarks (HellaSwag, ARC, MMLU, five more)

a filter helped if this curve rose at matched compute; nothing else counts

4 THE PIPELINE

Decanting the Web

The Pipeline



Extraction. trafilatura on the raw HTML, not the crawl's own text files.

Base filtering. URL blocklist, English at $p \geq 0.65$, Gopher's repetition rules.

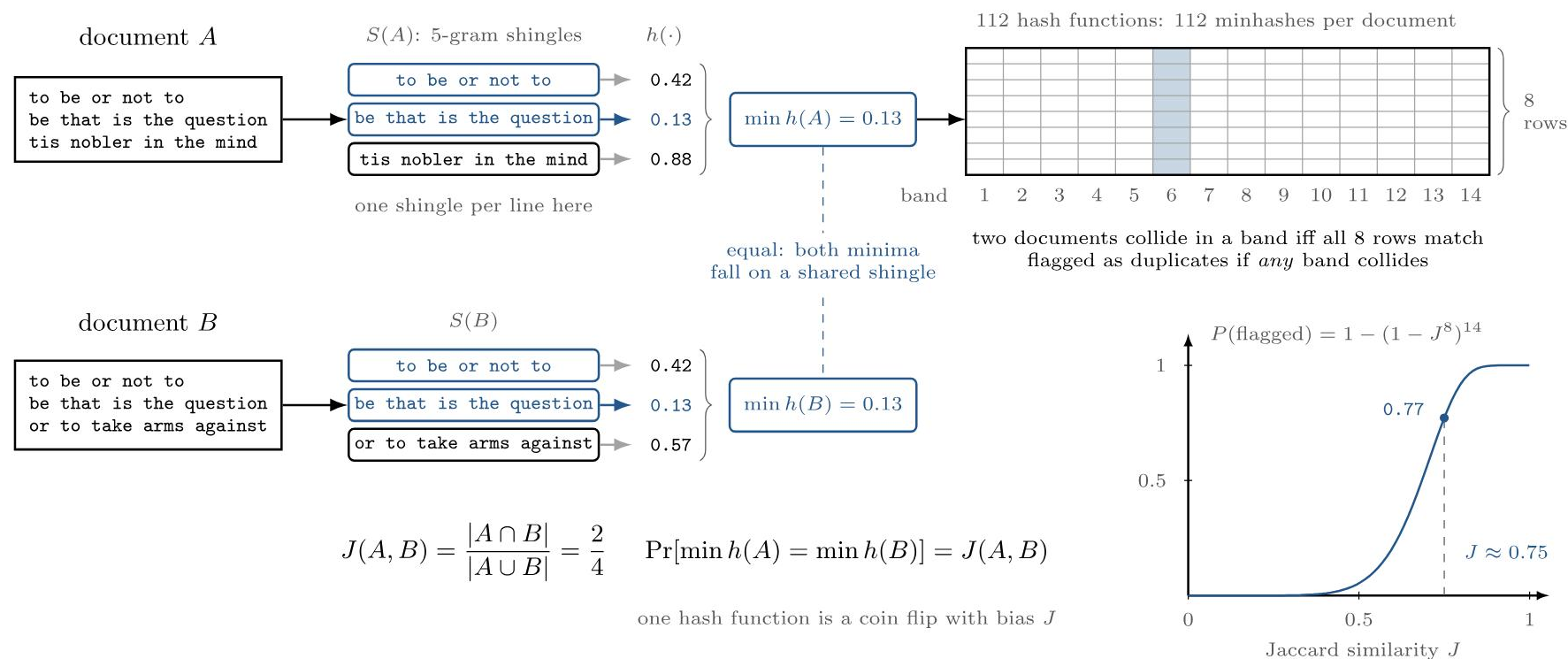
Deduplication. MinHash over 5-grams, inside each dump.

C4 and custom filters. Format rules, plus three FineWeb found itself.

Scrubbing. Emails and IP addresses anonymized for release.

Classifier. An educational score; keep 3 and up (FineWeb-Edu).

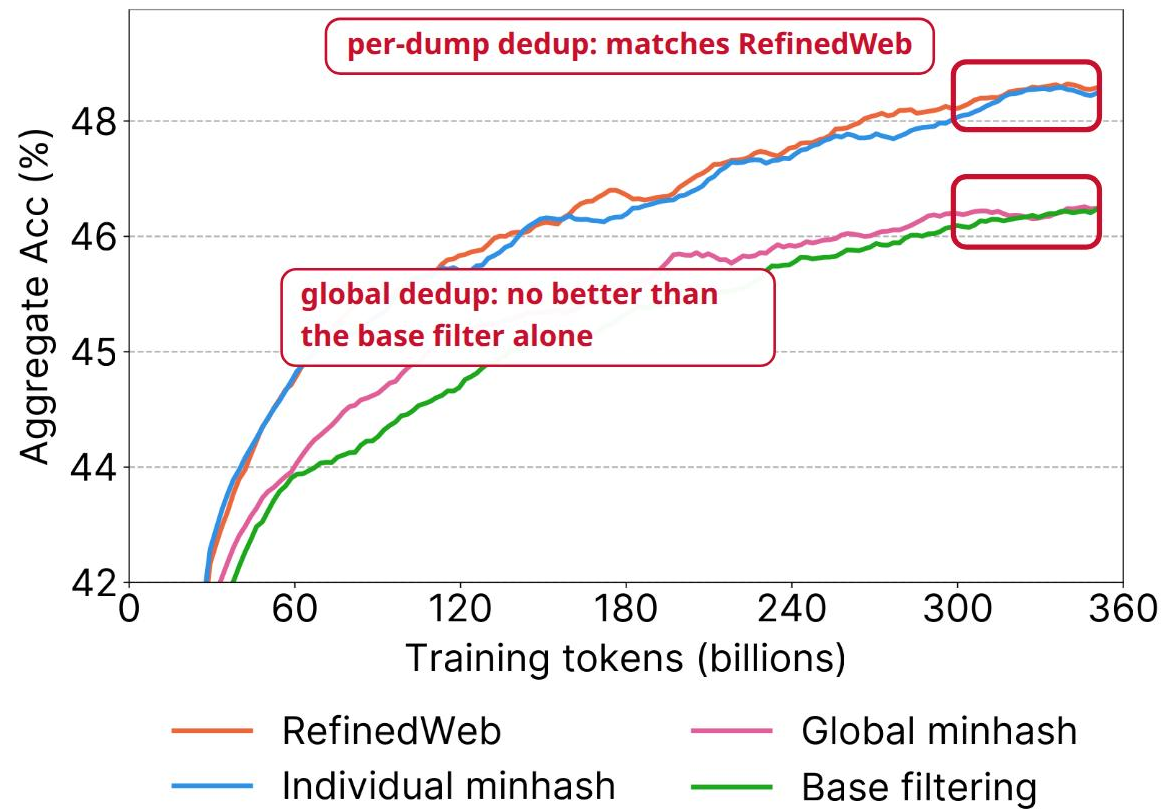
Deduplication



$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad \Pr[\min h(A) = \min h(B)] = J(A, B) \quad \Pr[\text{flag}] = 1 - (1 - J^8)^{14}$$

$J = 0.5$: flagged 5% of the time. $J = 0.75$: 77%. $J = 0.9$: 99.96%.

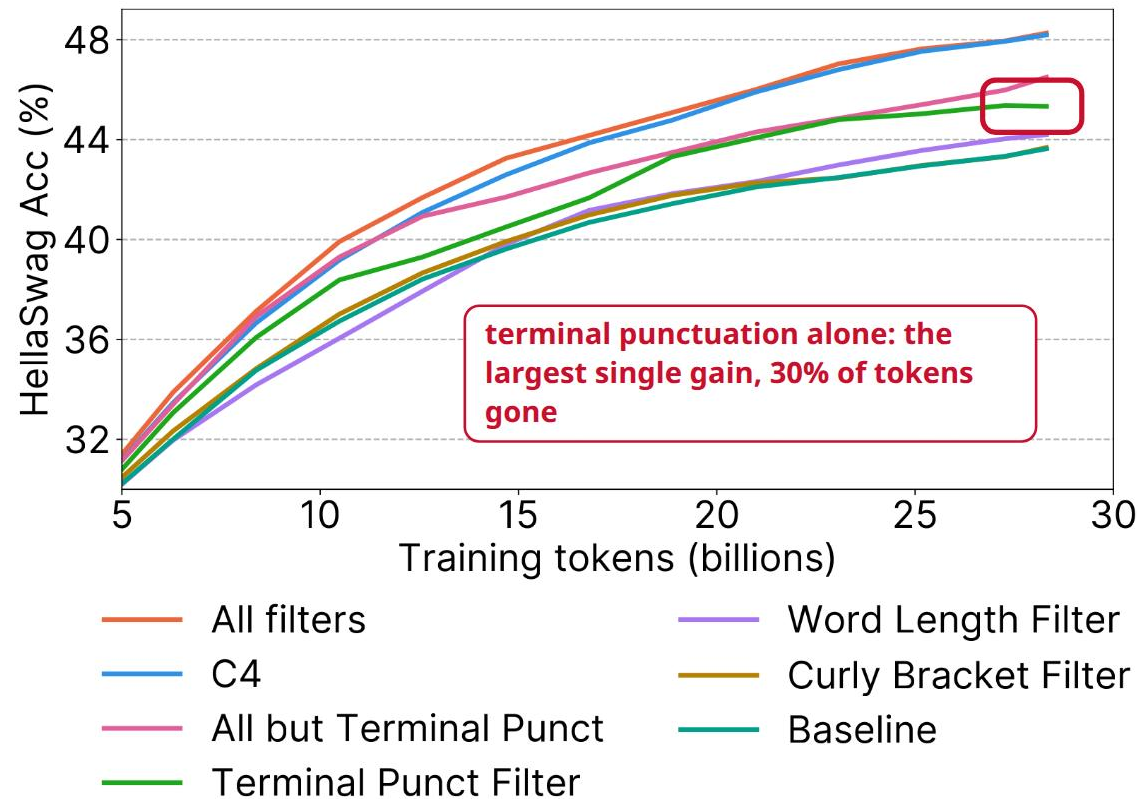
Deduplication



the experiment: MinHash across all 96 dumps at once, against MinHash inside each dump on its own

what global dedup did: 36T tokens down to 4T; on the oldest dump the 10% it kept scored worse than the 90% it removed

Heuristic Filters

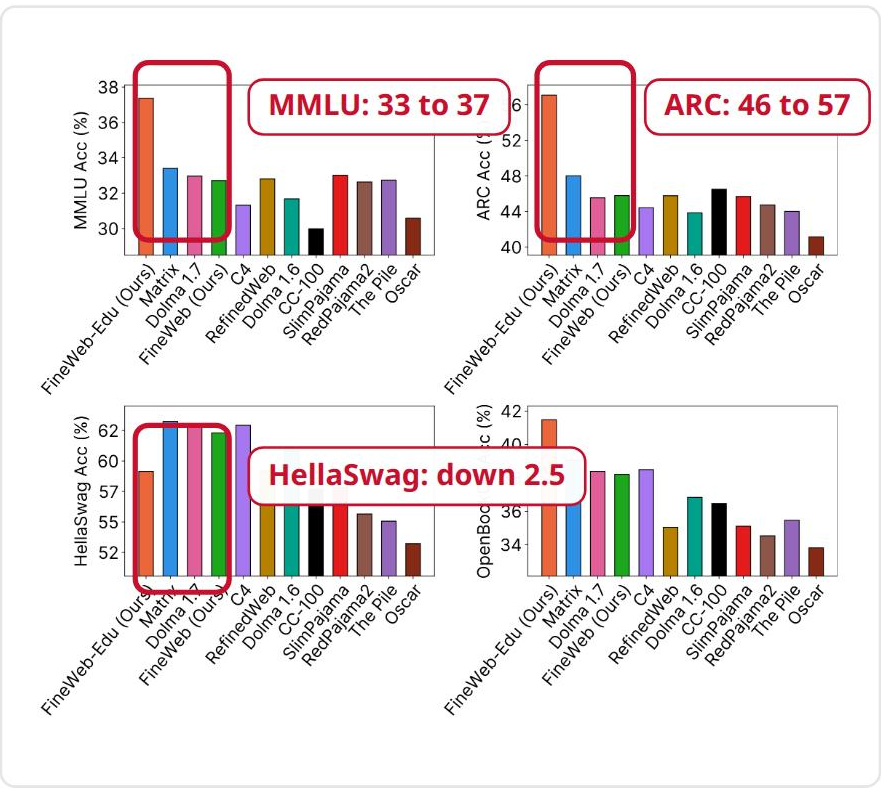
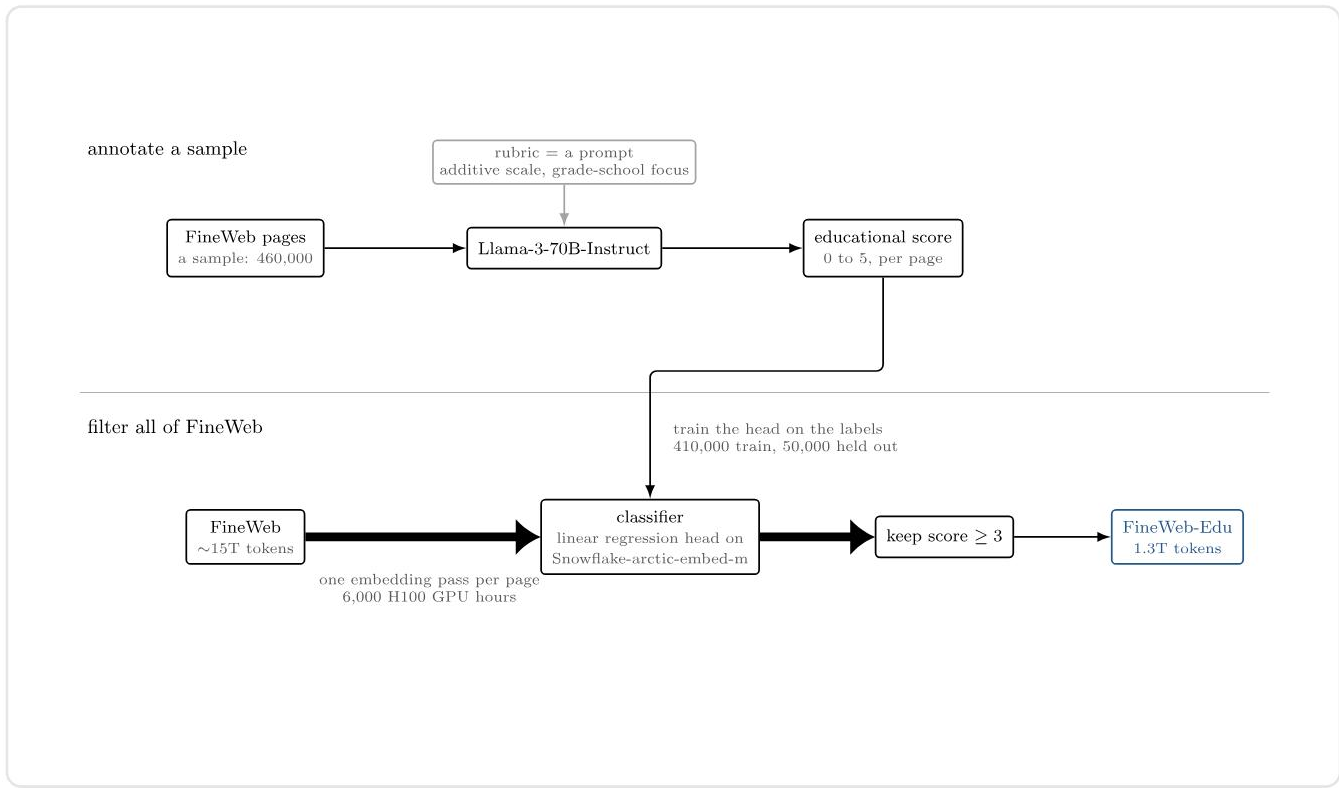


C4: lines end in punctuation, 5 words per line, 3 sentences per page, no lorem ipsum, no curly braces

Gopher: caps on repeated lines, paragraphs, and n-grams (30% of lines, 20% of characters)

FineWeb's own three: statistics that separate the good and bad sets of the last slide

A Learned Filter



annotator: Llama-3-70B-Instruct scores 460,000 pages 0 to 5 against a written rubric

classifier: a regression head on an embedding model, trained on those scores

threshold: keep score ≥ 3 ; 1.3T tokens survive of 15T

Keep or Drop

A `bigcreations.nl`

... to the online portfolio of BIG creations. Take a look around and get inspired! When a question rises about the possibilities or service, don't be afraid to contact by phone!

score 0.08 · dropped

B `abcnews.go.com/Health/texas-man-pioneers-beatless-heart`

Craig Lewis, a 55-year-old Texas man who was about 12 hours away from death, became the first human to receive what doctors are calling the "beatless heart."

score 2.76 · dropped

C `ni.com/documentation/en/daqexpress/3.2/node-ref/path-type`

Returns the type of the specified path, indicating whether it is an absolute, relative, or <Not A Path>. This node does not verify that the path exists on the computer.

score 2.62 · dropped

D `mathforum.org/library/drmath/sets/high_circles.html`

why 360 degrees? See also the Dr. Math FAQ: segments of circles. Browse High School Conic Sections, Circles. Selected answers: Find the center of a circle. Is a circle a polygon?

score 3.71 · kept

5 SYNTHESIS

Past the Data Wall

You Already Did This

Write a short story (3-5 paragraphs) which only uses very simple words that a 3 year old child would likely understand. The story should use the verb "decorate", the noun "thunder" and the adjective "ancient". The story should have the following features: the story should contain at least one dialogue, the story has a bad ending. Remember to only use simple words!

three required words, a toddler vocabulary, a required feature

Once upon a time, in an ancient house, there lived a girl named Lily. She loved to decorate her room with pretty things. One day, she found a big box in the attic. She opened it and saw many shiny decorations. Lily was very happy and decided to use them in her room.

As Lily was decorating her room, the sky outside became dark. There was a loud thunder sound, and Lily got scared. She ran to her mom and said, "Mommy, the thunder is so loud!" Her mom hugged her and said, "Don't worry, it will pass soon." But the thunder did not stop. It got louder and louder, and the ancient house started to shake. Suddenly, the roof fell down on the pretty decorations. Lily was sad because her room was not pretty anymore. The end.

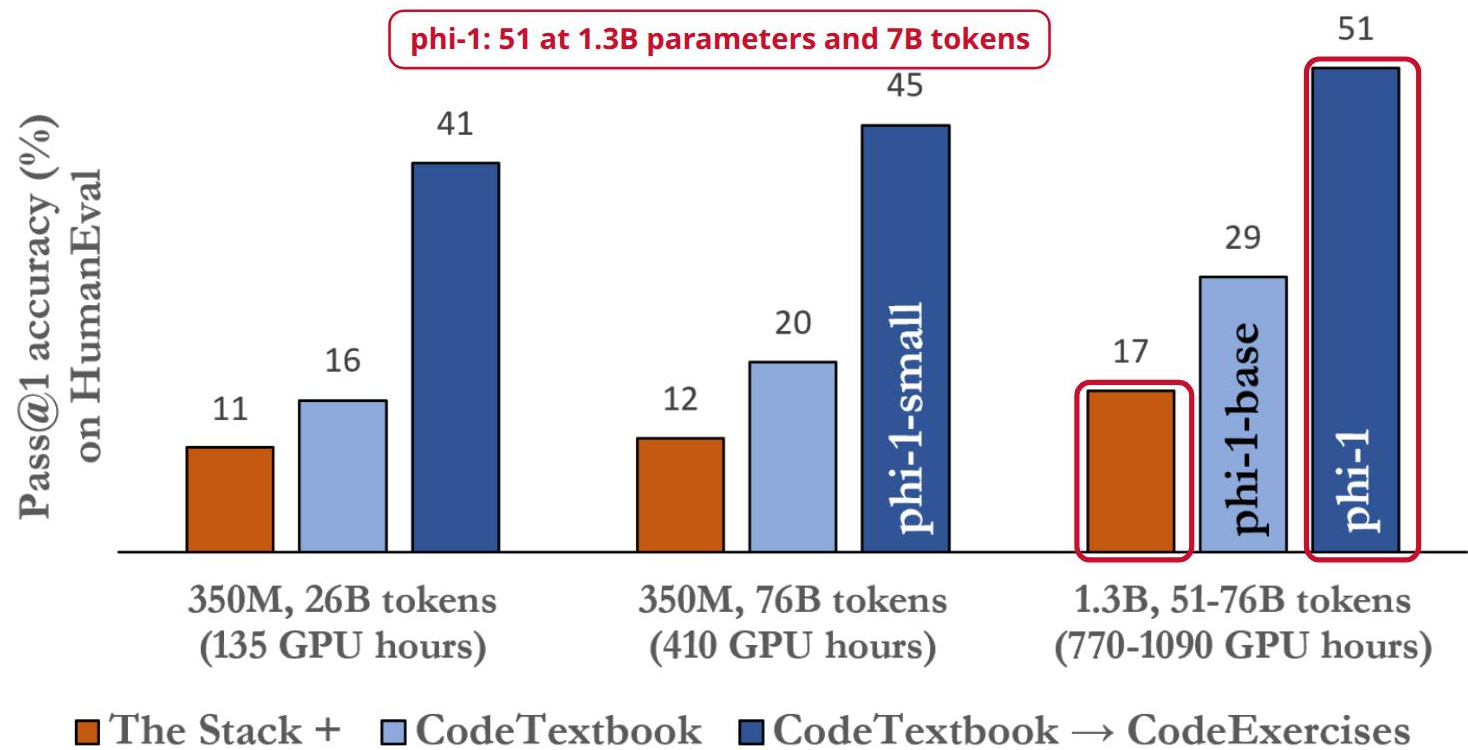
He asked Lily if she could play with Tom. Lily said yes and told her dad about the pond. She said yes and he put her toy car and her ball inside.

Tom and Lily were glad they had made her feel better.
your twin, 10.6M parameters, 82M characters of TinyStories

the authors: GPT-3.5 and GPT-4, from about 1,500 words a 3-year-old knows

your grid row: twin 0.70 on TinyStories, 2.62 on Shakespeare; base 1.43 and 2.13

Textbooks

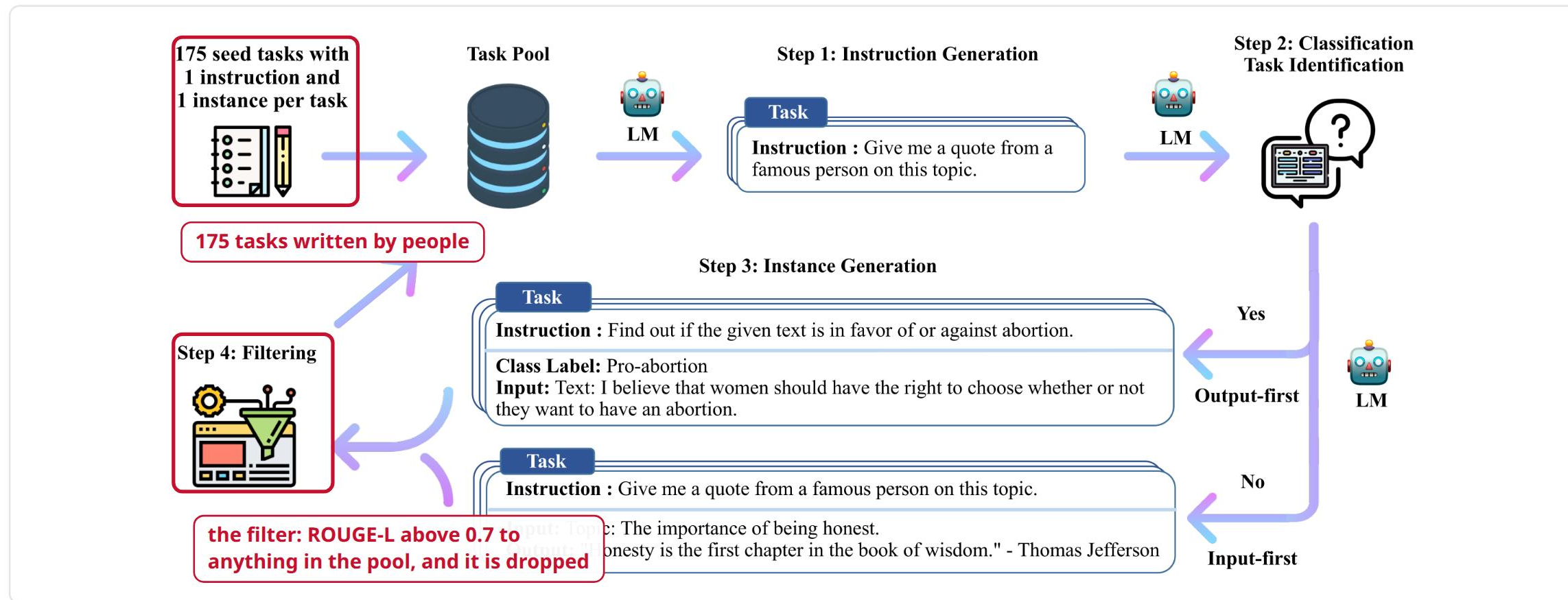


6B tokens of code: The Stack and StackOverflow, kept by a classifier trained on 100k GPT-4 quality labels

under 1B tokens of textbook: written by GPT-3.5 topic by topic, plus 180M tokens of exercises

for scale: StarCoder, 15.5B parameters on 1T tokens, scores 33.6 on the same test

Self-Instruct



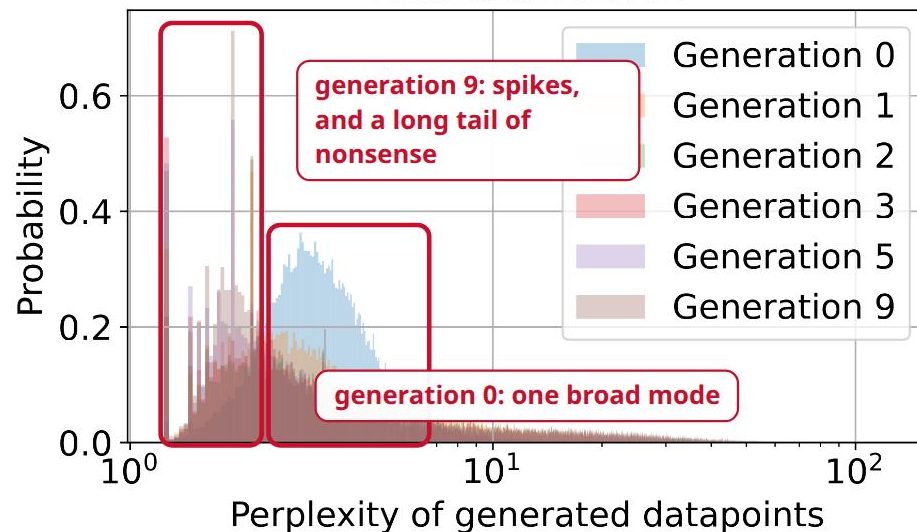
Evol-Instruct: rewrite each instruction harder: add a constraint, deepen, concretize (WizardLM)

Persona Hub: a billion web-mined personas as the axis of variation

at industrial scale: over 98% of Nemotron-4 340B's alignment data is synthetic

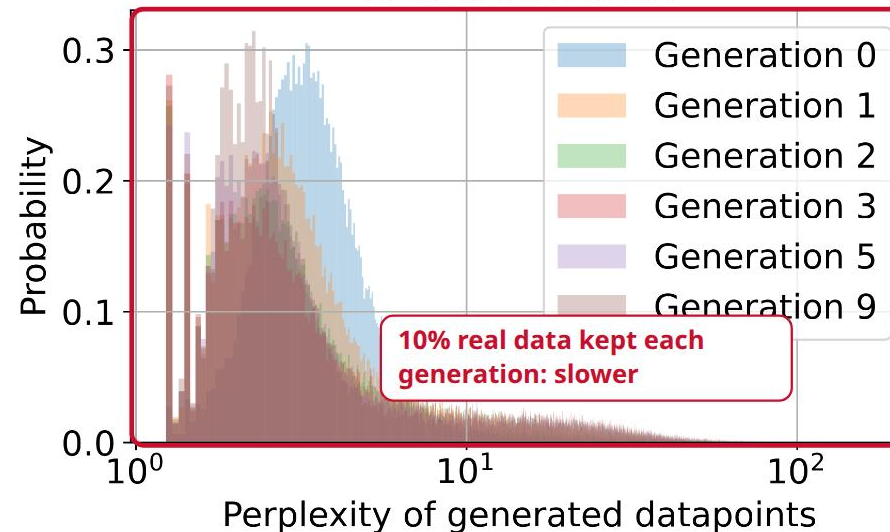
Collapse

Perplexity of generated datapoints
evaluated by model trained with
real wikitext2



(a) No data preserved

Perplexity of generated datapoints
evaluated by model trained with
real wikitext2



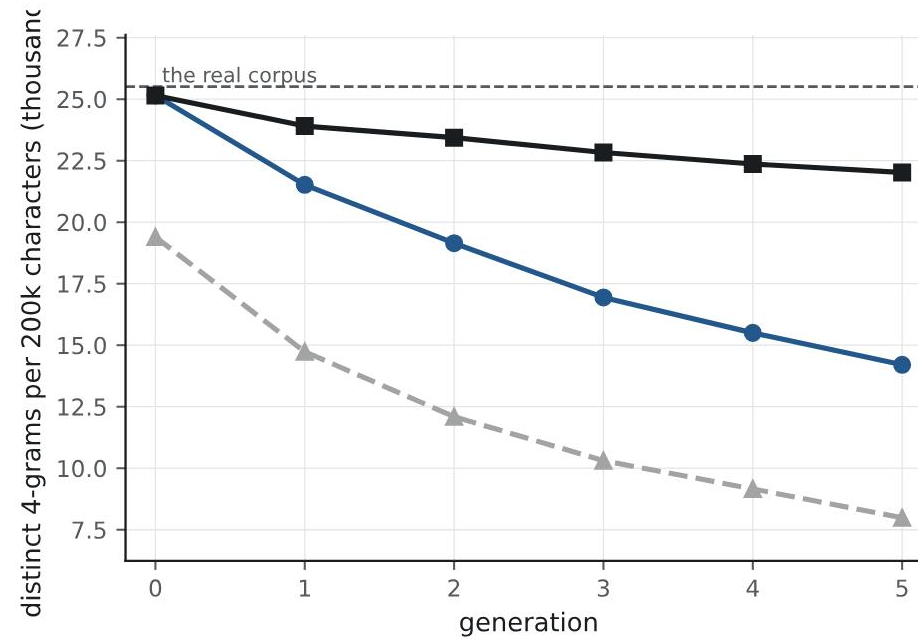
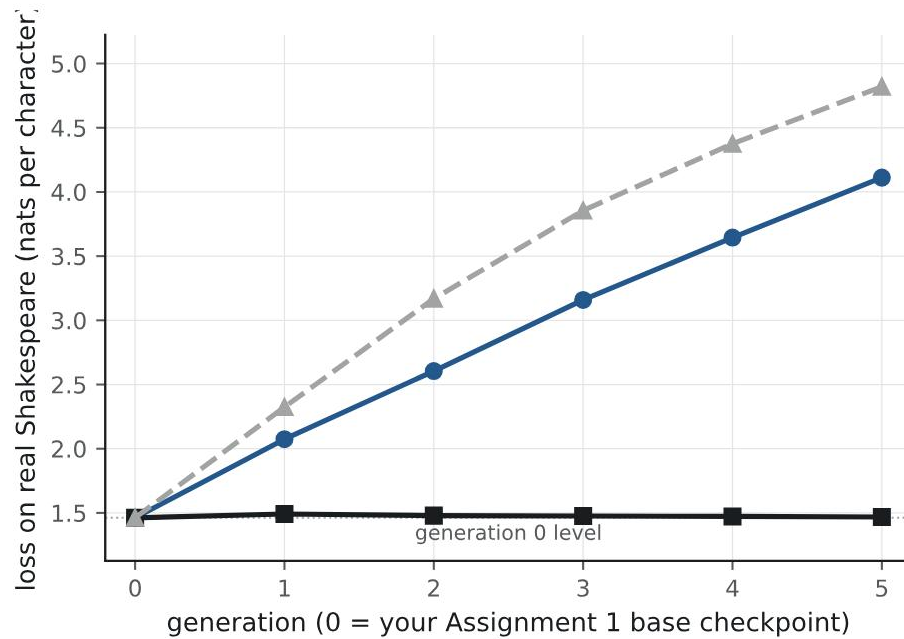
(b) 10% data preserved

statistical error: a finite sample never contains the rare events; the tails go first

approximation error: the model cannot represent the whole distribution, and errs the same way each time

recursion: generation k learns generation $k-1$'s mistakes as truth

Collapse

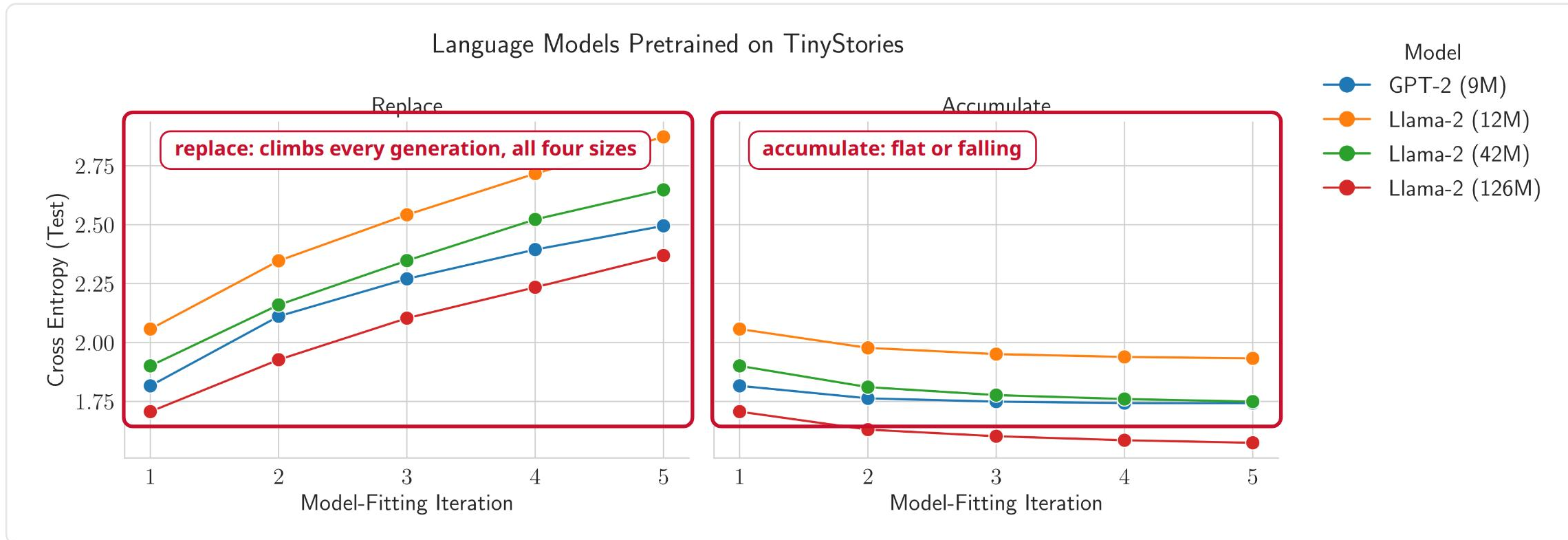


● replace: train on the previous model's sample only ■ accumulate: real text plus every sample so far ▲ replace, sampled at $T = 0.8$, top-k 40 (the A1 default)

the run: generation 0 is your out/base checkpoint; each generation samples 1.0M characters from the previous model and retrain from scratch; the exam is the real held-out split

MENENIUS:
Who has no more teet than you shall appear him honour with him?
First Citizen:
Ay, woman:
I fear the soul of words.
Second Citizen:

Accumulate



keep the real data: synthetic text joins the mixture; it never replaces it

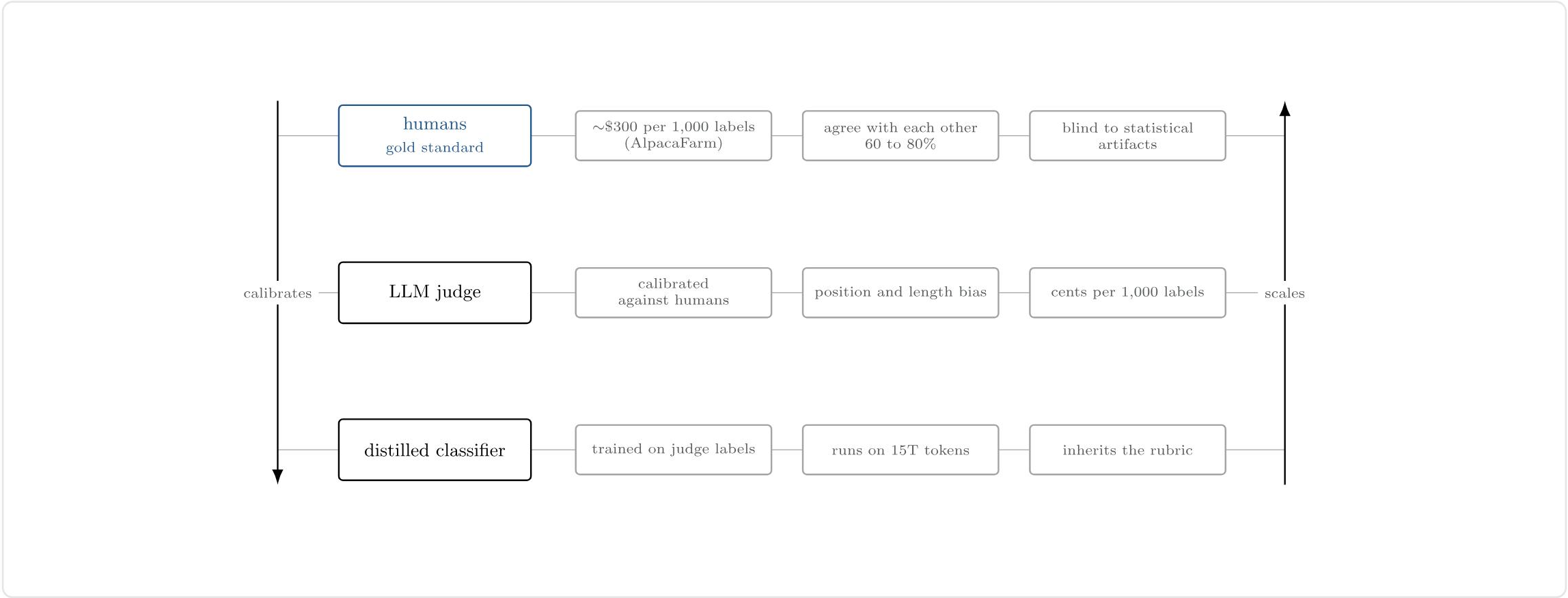
verify what you keep: tests, checkers, judges; a filtered sample is a different object from a raw one

tag provenance: know which tokens a model wrote, so the next crawl can tell

6 JUDGES IN THE PIPELINE

Who Grades the Data

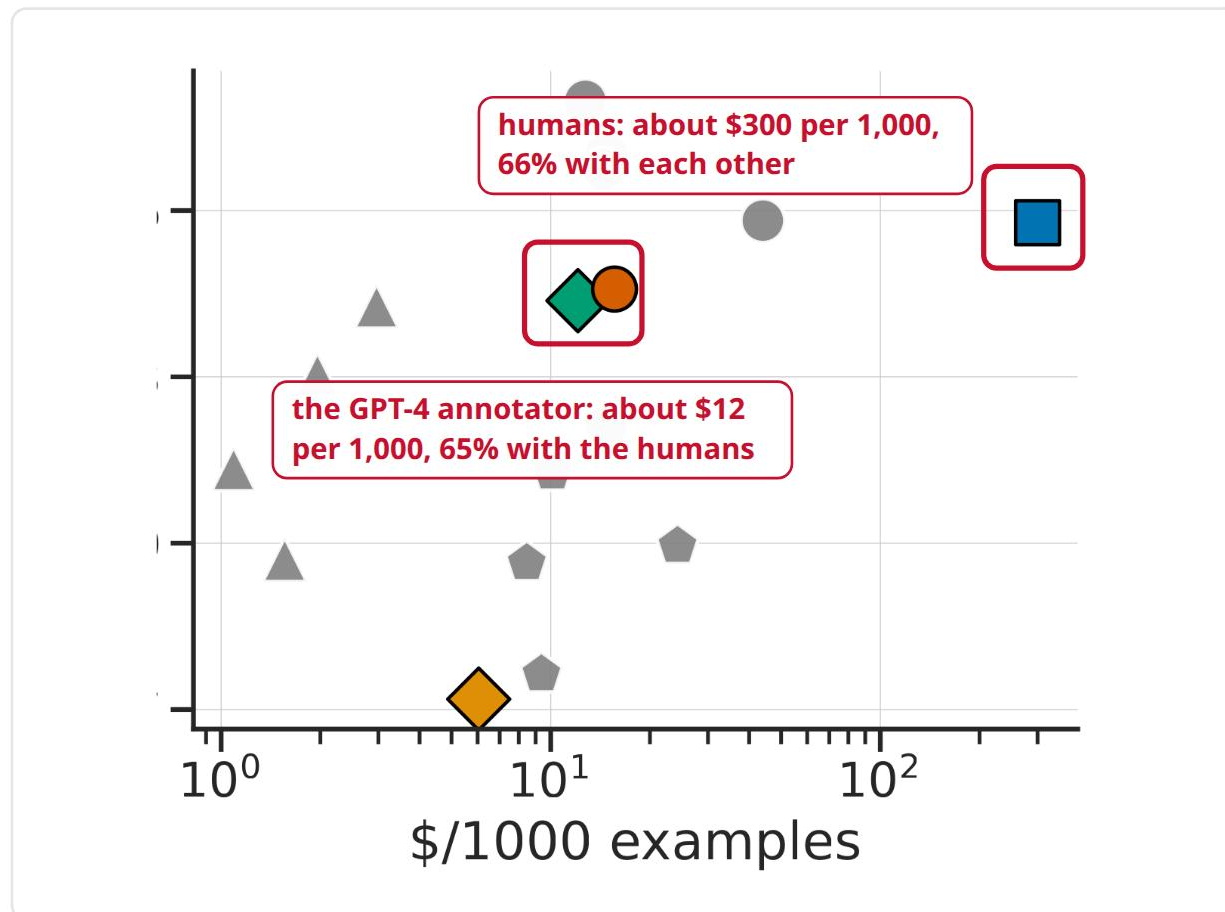
Three Meters



Humans set the meter, the judge reads it at scale, the classifier reads it on the whole web. Each inherits the biases above it.

LIMA: Zhou et al., 2023, arXiv:2305.11206. Judge agreement: Zheng et al., 2023, arXiv:2306.05685. Costs: Dubois et al., AlpacaFarm, 2023, arXiv:2305.14387. Classifier: Penedo et al., 2024, arXiv:2406.17557.

Calibrating the Judge



y: agreement with the majority of three human votes; x: dollars per 1,000 pairwise labels, log scale

the trade: 25x cheaper, at the human-human agreement level

the bias: the annotator prefers the longer answer 64% of the time; humans 62%; a policy trained against it learns to be long

Reading

Required

Penedo et al., The FineWeb Datasets, arXiv:2406.17557. The pipeline of section 2, with the ablation behind every stage.

Optional

Li et al., DataComp-LM (2406.11794). Xie et al., DoReMi (2305.10429). Wang et al., Self-Instruct (2212.10560). Shumailov et al., The Curse of Recursion (2305.17493). Gunasekar et al., Textbooks Are All You Need (2306.11644).

Assignment 2 is due Tuesday, September 15. The judge you run in Part 4, with its order swap, is the meter from today. Zheng and Miller from Thursday's list still stand as its reading.